Week 12 Group Project Report Guidelines (Stream 1)

MATH3888 (Interdisciplinary maths project unit)

Submission Deadline: **Friday, Week 12 (October 30th), 23:59**.

Submission and Formatting Instructions

Create your final group report using the LaTeX typesetting system. You are encouraged to use Overleaf, a collaborative online LaTeX editor, which you can access via your university account.

Submission of the final PDF file is via Canvas. One group member submits:

- **pdf**-file of the final project report.
- file name should be `groupXX-report.pdf` where `XX` is your two-digit group number.
- This written report is worth 20% of your final mark.

LaTeX Setup and Constraints

Use the following basic setup for your LaTeX file:

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath}
\begin{document}
...
\end{document}
```

- **Font Size & Margins:** The font size is strictly 11pt. The `geometry` package will set the margins as required.
- **Packages:** The `amsmath` package is recommended for mathematical typesetting. You may add other packages as needed.
- **Page Limits (Adjusted for Group Size):** To ensure fair workload distribution, maximum page limits (excluding the reference list) are strictly enforced based on your group size:
  - **Standard Groups (7–9 members):** Maximum 12–15 pages.
  - **Small Groups (5 members):** Maximum 8–10 pages.

GUIDELINES FOR THE STRUCTURE OF YOUR REPORT

Title and Group Roster

Provide a succinct title that describes your project, ideally fitting on one or two lines. Immediately below the title, include your Group Number and a **Contribution Roster** (mapping to your Week 5/8 project pitch) explicitly stating which sections each group member was primarily responsible for.

Part 1: The Biological Setting (Introduction)

This section serves as your general introduction to the biological system.

- Describe the main biological features of the cell under study relevant to your project. Cover the relevant literature (restrict to standard textbooks and original literature in peer-reviewed journals).
- Detail the cell's signaling system. You *must* include a diagram of the signaling pathway(s) you are considering.
- Conclude this section with a clear and exact statement of your group's study aims.

Part 2: Agonists, Drugs, and Experimental Data

- Introduce the agonists, enhancers, drugs, or blockers used in your experiments. For each substance, state your hypothesis regarding its mechanism of action within your biological system.
- Describe your group's chosen experimental protocols in detail so the context of the data is clear.
- Present all experimental data and findings for the main agonist(s) and drug(s). Use clearly labeled tables, graphs, and figures, including relevant time traces (e.g., calcium, voltage). Do not duplicate data between text and figures.

Part 3: Mathematical Modeling & Hypotheses (Methods)

This section details how you mathematically translated the biological system.

- Present your ODE system for the cell signaling pathway. Explicitly explain its relationship to the biological system, detailing how key parameters and functions correspond to specific biological components.
- Translate your biological hypotheses about the agonists/drugs into specific mathematical hypotheses within your model. What parameter changes represent the drug interventions?

Part 4: Computational Modeling Results

Simply state and show what you found computationally. Do not interpret the biological implications here.

- **Baseline dynamics (Python):** Provide the key findings from your base mathematical model, showing numerical output time-series plots that capture the required dynamics.
- **Advanced analysis (`PyDSTool`):** Present your bifurcation analysis with respect to your chosen parameter.
- Ensure all numerical output includes appropriate legends, captions, and axis labels that enable them to be interpreted without extensive reference to the text. Briefly highlight the key mathematical or computational tools you used.

Part 5: Interpretation and Discussion

This section is about the interpretation of your combined results.

- Compare your experimental and computational modeling results. Did the substances behave as you anticipated in both settings? If not, suggest why not.
- Discuss strengths and limitations of the mathematical approach you took. Address any technical bottlenecks (e.g., stiffness issues) encountered during modeling and how your group pivoted.
- Place your findings into the context of the overall biological system. Suggest future modeling studies or extensions that could build upon your work.

Reference List

- Provide a complete, properly formatted list of all sources and literature cited in your report. Ideally, use BibTeX with the `natbib`-style.

Grading Rubric

Criteria	Excellent (HD)	Very Good (D)	Good (CR)	OK (P)	Poor (F)
Knowledge of discipline area	Commanding breadth and depth of knowledge of the biological system and mathematical theory. Demonstrates very strong understanding of the project's context and implications.	Comprehensive depth of knowledge with a strong focus on the specific research field, with some appreciation of the broader implications.	Sound knowledge of the immediate area of study, with some appreciation of the broader biological and mathematical implications.	Satisfactory knowledge of the immediate area of study, with limited appreciation of the broader context.	Poor or limited knowledge of the discipline area, and lack of understanding of the context of the work.
Inter-disciplinary Integration	Extremely creative and appropriate translation of biology into mathematics. Flawless synthesis of experimental data and modeling results.	Appropriate and accurate translation of biology into mathematics with some elements of creativity. Clear links between data and the model.	Appropriate translation of biology into mathematics. Satisfactory links between the experimental data and the mathematical model.	Satisfactory attempt to model the biology. Attempts to link the mathematical output to the biological hypotheses.	Missing or poor integration. No obvious attempts to link the biological hypotheses to the mathematical findings.
Report style, tables, figures	Highly cohesive, concise, and logical report. All figures/tables have legends and captions that can be interpreted without referencing the text.	Reasonably cohesive and logical. Some figure legends and captions may lack sufficient detail.	Report is overall satisfactory but some sections should display better logic or flow. Figure legends and captions may be poorly explained.	Report is satisfactory but lacks group cohesion (reads as fragmented parts). Figure legends and captions may be poorly explained.	Inadequate level of detail, cohesion, and/or logic. Missing figure legends and captions.